

Texture Semantics is Robust to Scaling

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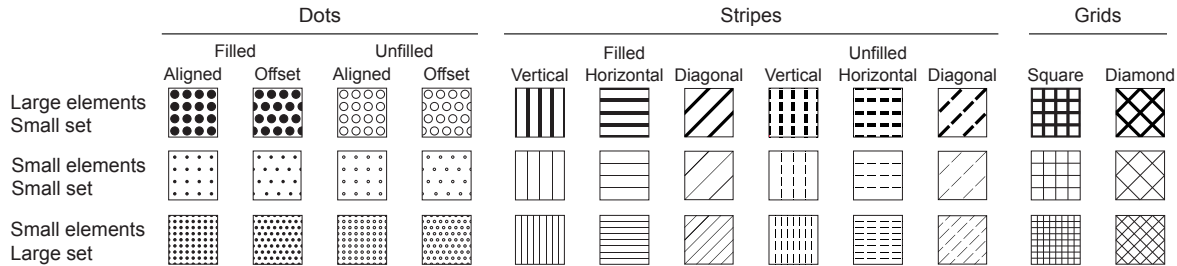


Figure 1: The 36 ‘original’ textures (adapted from [4]) include three element types (dots, stripes, grids) in three versions of varying element size and set size. The dots varied in alignment (aligned, offset) and fill (filled, unfilled); the stripes varied in orientation (vertical, horizontal, diagonal) and a variant of fill (solid, dashed); and the grids varied in shape (square or diamond).

ABSTRACT

Studies of visual semantics for information visualization aim to understand observers’ expectations about the meaning of visual features (e.g., color, texture) because visualizations that align with those expectations are easier to interpret. Previous work on visual semantics focused primarily on color, with the implicit assumption that color semantics is unaffected by changes in the size of the visualization (given sufficient perceptual discriminability across sizes). Changing size from small scale (e.g., small figures in a paper) to large scale (e.g., large figures in a slide presentation) is straightforward for visualizations that have solid colored regions, but can be more complicated for visualizations with heterogeneous textures because there are multiple ways to scale textures—zooming or repeating texture elements. Previous work suggested that original textures were more perceptually similar to repeat-scaled rather than zoom-scaled textures. Here, we found that texture semantics was preserved after both types of enlargement, suggesting that texture semantics is robust to scaling, at least for geometric textures in which elements are visible at all scales.

Index Terms: information visualization, perceptual semantics, visual reasoning, texture perception, visual communication

1 INTRODUCTION

Research on *visual semantics* for information visualization aims to understand the meanings observers ascribe to visual features. This understanding can help design visualizations that align with observers’ expectations about feature meaning, and thus facilitate visual communication. Research in this field has primarily focused on color, studying how to optimize color encoding systems for semantics [6, 14, 12, 7], quantify the effects of optimized design on interpretability [6, 12, 11, 7, 13], and predict conditions under which it is possible to meaningfully represent concepts using color [7].

An implicit assumption in these studies is that color semantics is consistent regardless of the size of the colored regions, as long as those colors are perceptually discriminable (see [17] and [18] for discussion on perceptual discriminability of colors given mark

size/shape). For example, it is assumed that the semantics of a small colored bar chart in a paper will hold when it is enlarged on a slide in a presentation, assuming faithful color rendering by the display.

When changing the size of figures with homogeneous, solid colored regions, the notion of scaling seems simple—just proportionately increase/decrease the size of the figure (e.g., click a corner of a figure in, say, PowerPoint and drag it to be larger/smaller). We refer to this approach as ‘zoom-scaling.’ However, when changing the size of non-homogeneous, patterned regions, as in visualizations that encode concepts using texture¹, there are at least two ways to scale: zoom-scaling and ‘repeat-scaling.’ Repeat-scaling is a non-proportional scaling that increases size by repeating the pattern of elements over space while preserving element size [5, 2].

For example, Figure 2A shows an original texture that is either repeat-scaled by repeating the dots in the same size and spacing or zoom-scaled by proportionately enlarging the original.² Previous studies found that when observers were asked to report whether an original image composed of many small elements appeared more similar to a zoom- or repeat-scaled image, they chose the repeat-scaled image [5, 2]. This finding implies that preserving *appearance* when enlarging visualizations constructed from textures of many elements may require a more effortful redesign, which repeats texture elements in their original size (Figure 2B, left) rather than simply zooming in on the whole image (Figure 2B, right).

In this study, we tested the hypothesis that texture *semantics* may also be better preserved after repeat-scaling compared to zoom-scaling, at least when there were many elements in the textures. If so, that would be further reason to consider doing a more effortful redesign that repeats texture elements when enlarging visualizations. A reader may suppose that using a legend to explicitly specify the mapping between textures and concepts in a visualization would protect against changes in semantics due to scaling. But, if semantics changes and does not match the mapping in the legend, the visualization will be harder to interpret [6, 10, 15, 16].

Contributions. Our study provides evidence that texture semantics is preserved after both types of scaling, contrary to our hypothesis. These results suggest that texture semantics is robust to scaling, at least for the concepts and geometric textures in the sizes tested

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¹We use the term ‘texture’ following previous literature (e.g., [1, 5, 4]), but see [3] for considerations on using ‘pattern’ instead.

²A third type of scaling is a hybrid between zoom- and repeat-scaling, which zooms the spacing but retains the stroke weight from the original. This intermediate case was not studied in [5, 2] or in the present work.

in this study. We also assessed which kinds of geometric textures are semantically discriminable and provide evidence that semantic discriminability relies on varying element shape (dots/stripes/grids) for the concepts we studied.

2 BACKGROUND ON ASSESSING VISUAL SEMANTICS

Our approach to studying texture semantics was motivated by prior work on assignment inference for color semantics [7, 9, 12], see Schloss [9] for a review. Assignment inference is the process observers use to infer which perceptual features correspond to concepts in visual encoding systems [12], such as which colors mean which categories in a bar chart [7].

Evidence suggests that observers perform assignment inference by comparing the possible assignments in an encoding system, and inferring the assignment that maximizes the system's total goodness or 'merit'. For example, Figure 3 shows bipartite graphs for encoding systems corresponding to visualizations representing data for the concepts mango (M) and watermelon (W), studied in [11]. Each encoding system contains two concepts (circles) and two colors (squares), where the thickness of the edges connecting them represents the merit of each pairing. Here, we treat merit as the association strength between each individual color and concept, but merit is more complex for encoding systems with > 2 concepts [12, 7]. Assignment inference operates by comparing the possible assignments and returns the assignment that has the greatest total merit; the *optimal assignment* [12]. For example, the encoding system in Figure 3A has two possible assignments, the 'outer assignment' that connects M-O and W-R (i.e., outer edges of the bipartite graph) and a 'inner assignment' that connects M-R and W-O (i.e., inner, crossed edges of the bipartite graph).³ Here, the optimal assignment is M-B/W-R, which lead observers to infer that orange means mango and red means watermelon [11].⁴

The ability to perform assignment inference for interpreting visualizations depends on *semantic discriminability* of the perceptual features, or the ability to tell them apart in terms of their meaning [11]. Semantic discriminability is based on the degree to which one assignment has greater overall merit than the alternative(s). In Figure 3A, assignment inference is easy because the outer assignment has far greater merit than the inner assignment. However, in Figure 3B, assignment inference is harder because both assignments have similar merit. Semantic discriminability can be quantified using **semantic distance** (ΔS), a metric that computes the probability of one assignment over the alternative while accounting for uncertainty in the encoding system (see Supplementary Material Section S.1 for further description and a formal definition). As semantic distance increases, observers' ability to perform assignment inference improves [11, 7]. Thus, we can use semantic distance as a proxy for performance in assignment inference.

3 EXPERIMENT

In the present study, we tested the hypothesis that texture semantics would be better preserved after repeat-scaling, at least when textures were made of many texture elements. Our hypothesis was motivated by examples such as in Figure 2B-C, where the bars in Figure 2B are imagined to represent the concepts strawberry (S) and watermelon (W). The corresponding bipartite graphs in Figure 2C illustrate a potential scenario in which the inferred meanings of textures in the original visualization are preserved after repeat-scaling but reverse after zoom-scaling. In this hypothetical example, we

³Which assignment is 'outer' vs. 'inner' is arbitrary and does not pertain to encoding systems for > 2 concepts, but the terms are useful in referring to which assignment is optimal in encoding systems for 2 concepts.

⁴Here, the optimal assignment maps colors to their most strongly associated concepts, but sometimes assignment inference maps colors to weakly associated concepts to optimize the whole system (global assignment) [12].

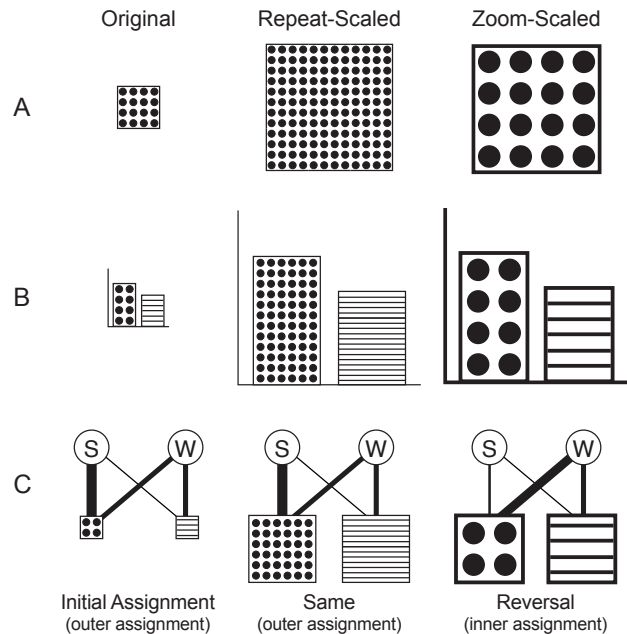


Figure 2: (A) Example of an original, repeat-scaled, and zoom-scaled texture, with (B) corresponding example bar charts for each type of scaling and (C) a hypothetical example of how inferred mappings could reverse after zoom-scaling but not repeat-scaling.

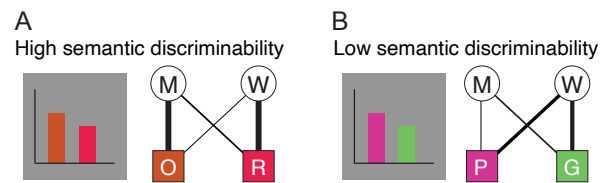


Figure 3: Charts of data representing Mango (M) and Watermelon (W) with bipartite graphs showing the encoding system between colors and concepts for each chart. The encoding systems have (A) high semantic discriminability or (B) low semantic discriminability (figure adapted from [11, 9]). In both examples, the 'outer assignment' is the optimal assignment because it has greater total merit.

reasoned that in the original size, strawberry would be strongly associated with dots and watermelon would be associated similarly with dots and stripes so the optimal assignment would map strawberry to dots and watermelon to stripes (outer assignment). This assignment would be the same for repeat-scaled textures because they are perceptually similar to the originals. But, the assignment could reverse for zoom-scaled textures because the large black dots would be especially associated with watermelon seeds, thus giving greater merit to the inner assignment.

To test for reversals due to scaling, we used the following approach. First, we collected texture-concept association ratings for textures in an original size (Figure 1), zoom-scale versions, and repeat-scale versions (Section 3.1). We then used these data to determine the semantics for each pair of textures in their original size, with respect to each pair of concepts. We operationalized texture semantics as the optimal assignments for each texture and concept pair (i.e., which texture concept pairing has greater merit; Section 2). Then we determined the optimal assignment for zoom-scale and repeat-scale versions of those textures.

Prior to testing for reversals, we thresholded the texture pairs for each concept pair to only include those that had high semantic dis-

tance in their original size. This way, we could test for reversals within encoding systems in which concepts were strongly differentiated by textures. Moreover, textures of low semantic distance would be less advisable to use in the original size design because they are less interpretable and (B) reversals when semantic distance is low should have little impact on interpretability. We initially set the semantic distance threshold to $\Delta S = 0.7$ based on data from Schloss et al. [11] suggesting this ΔS should produce interpretation accuracy of 70% (see Supplementary Material Section S.2 for details). However, we note that our key findings persist for far lower thresholds of ΔS (see Section 3.2).

Finally, we tested for reversals in assignments between the original version and each of the scaled versions of each texture pair. A reversal implies that the inferred meaning of textures in their original size changes to the opposite meaning after scaling. If reversals exist, that would mean that an encoding system that is optimized to be maximally intuitive in a small scale could be maximally confusing at a larger scale if the designer maintains the same texture-concept mappings after scaling. In contrast, a lack of reversals would suggest that texture semantics is robust to scaling, at least for the encoding systems studied here.

The stimuli, data, and analysis code can be found at OSF⁵.

3.1 Methods

Participants. 51 undergraduates participated for extra credit in their Psychology course (M age = 18.6 years). Their free-response reported demographics were as follows. Gender: 35 women; 16 men. Race/ethnicity: 37 White, 2 Black or African American, 12 Asian, 3 Hispanic or Latinx, and 1 “mixed”.⁶ All reported normal/corrected to normal visual acuity. All participants gave informed consent and the UW–Madison IRB approved the protocol. We analyzed data from $n = 45$ because output was missing from 6 participants due to experimenter error.

Design and Displays. The participants rated the association between each of 180 textures for each of 5 concepts (540 trials) in a randomized block design; all participants judged all textures for a given concept in a random order before advancing to the next concept. Concept order was also randomized across participants. The stimuli were presented on a 24.1-in. ASUS ProArt PA249Q monitor (1920 × 1200 resolution). We controlled viewing distance (65 in.) by having participants center their face against a window (12 in. × 7 in.) made of black fabric suspended across the testing booth.

Textures. The textures included the 36 ‘original’ textures in Figure 1 (1×1 in.), 36 repeat-scale versions (3×3 in.) and 36 zoom-scale versions (3×3 in.). The 36 textures were inspired by the textures in [4], but were modified to systematically vary element shape, fill, density, orientation, and size. They were also designed so the original textures would appear continuous when tiled to produce the repeat-scale versions. That is, when aligned edge-to-edge, the patterns seamlessly extended across adjacent textures, creating the appearance of a single, continuous surface. The repeat-scale versions were produced by repeating the original size textures in a 3×3 grid. The zoom-scale versions were generated by resizing the dimensions of the original size figure to match the overall size of the repeat-scale texture (i.e., enlarging the elements proportionally).

Concepts. Each texture was rated for each of five concepts: strawberry, carrot, watermelon, mushroom, and corn. We chose to test concrete concepts referring to objects with observable shapes and textures to help ensure participants would have systematic texture concept associations. We chose these specific fruits/vegetables because they vary in shape/textures: strawberries are dotted; carrots are stripe-like; watermelon has dots inside and stripes outside; and mushrooms are dot-like with stripe-like fibers, and corn kernels are dot-like, while kernels on cobs are grid-like and cobs are stripe-like.

⁵<https://osf.io/s7ghe/>

⁶Some participants reported ≥ 1 racial/ethnic identity.

Table 1: Correlations of associations between original and zoom-scale (r_{zoom}) or repeat-scale (r_{repeat}) textures for each concept with Fisher’s z and p values used to compare correlations.

Concept	r_{zoom}	r_{repeat}	z	p
strawberry	.99	.98	1.42	.16
watermelon	.94	.93	0.32	.75
carrot	.98	.99	1.42	.16
mushroom	.87	.87	0.57	.57
corn	.95	.93	0.70	.48

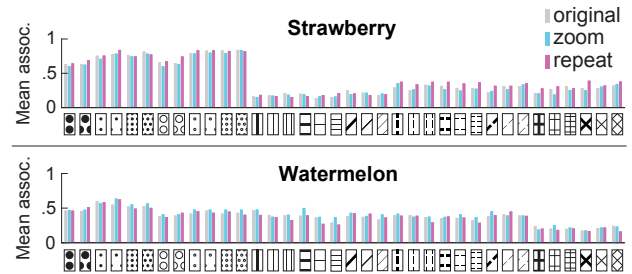


Figure 4: Mean association ratings for the concepts strawberry and watermelon for the 36 textures (x-axis) in their original (gray bars), zoom-scale (cyan bars), and repeat-scale (magenta bars) versions. See Supplementary Figure S.4 for the full set of concepts.

Variability in associations across concepts should result in semantically discriminable textures (semantic discriminability theory [7]). Although our rationale emphasizes resemblance between textures and concepts, previous work on color found that visual semantics works similarly for concrete and abstract concepts [7].

Procedure. The participants were told they would be presented with textured squares that would vary in size, one at a time, and they would be asked to rate how much they associated each texture with the concepts strawberry, carrot, watermelon, corn, or mushroom. They were instructed to make their rating on a scale from ‘not at all’ to ‘very much’ by sliding the cursor along the response scale and clicking to record their response (Figure S.2). They were also told there would be five blocks of trials, one for each concept, and they would be asked to rate all textures for each concept before going on to the next block. Then, they were shown all 36 textures (original size) with all five concepts and were asked to point to the textures they associated most/least with each concept. This anchoring was done to help participants know what associating ‘not at all’ and ‘very much’ meant for them in the context of these textures and concepts [8]. During the experiment, each block began with a display indicating which concept they would be asked to rate during that block. They then proceeded to the 108 trials for that block, with each trial separated by a 500 ms. inter-trial interval.

3.2 Results and Discussion

We first assessed how similar the texture-concept associations were for the original textures compared to each type of scaled texture. As shown in Figure 4 for two example concepts, the pattern of associations was highly similar for all three texture versions (see Supplementary Figure S.4 for the full set of concepts). For all concepts, mean associations were highly correlated between original and zoom-/repeat-scaled textures, with no significant difference in correlations between scaling methods (Table 1). Thus, associations between individual textures and concepts were similar after scaling.

This correlation analysis concerned associations between each texture and concept in isolation, but visual semantics depends not just on isolated associations; it depends on the relative associa-

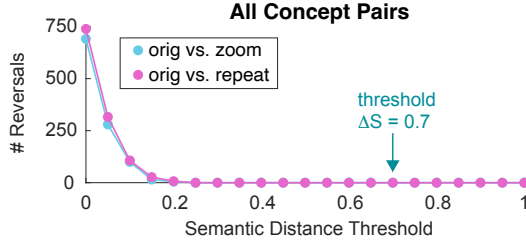


Figure 5: Number of assignment reversals after zoom-scaling (cyan) or repeat-scaling (magenta) as a function of semantic distance ΔS threshold (combined over all concept pairs). [Note: There were 6,300 possible reversals (630 unique texture pairs $\times 10$ concept pairs).]

tions within an encoding system ([12, 9]. Moreover, despite the high pre-/post-scaling correlations, there could be particular texture pairs that had different enough associations after scaling to produce assignment reversals. To test for changes in texture semantics, it is necessary to evaluate the optimal assignments, given all textures/concepts within encoding systems.

As motivated in the beginning of Section 3, we computed ΔS for each pair of textures within each pair of concepts and determined which texture pairs passed the $\Delta S \geq .7$ threshold for each concept pair. We then determined the optimal assignments for each of those texture/concept pairs (i.e., outer or inner assignment) for the original, zoom-, and repeat-scale textures. Finally, we counted the total number of reversals after each type of scaling. We were surprised to find zero reversals for texture pairs that passed the $\Delta S \geq .7$ threshold, suggesting that pairs of concepts that are well-differentiated via texture remain so, regardless of how they are scaled. Concerned this threshold was too conservative to detect reversals, we tried varying the threshold from 0 to 1 in increments of .05 and counted the number of reversals at each ΔS threshold (Figure 5). Over all concept pairs, there were no reversals for zoom- or repeat-scale textures until the threshold decreased to $\Delta S = 0.2$. This ΔS is so low that observers are unlikely to have systematic expectations of texture meaning (i.e., performance in assignment inference near chance) [11] so reversals would have little consequence for interpretability.

Finally, we examined which kinds of textures support high semantic discriminability above the threshold of $\Delta S \geq .7$. For some concept pairs no textures passed the threshold, which is likely because they had similar texture-concept association distributions (see Supplementary Material Figure S.4; semantic discriminability theory [7]). For concept pairs that did have texture pairs that met the threshold, the textures always combined different kinds of elements (dots, stripes, or grids). That is, in Figure 6, no texture pairs that were along the diagonal sets (i.e., dots with dots, stripes with stripes, or grids with grids) passed the threshold. Texture pairs that differ in element fill, density, size, and orientation but contain the same element shape did not have sufficient semantic discriminability (Figure 6). These results suggest that creating semantically discriminable textures relies on varying element shape, at least for the concepts studied here.

4 CONCLUSIONS AND OPEN QUESTIONS

We initially aimed to test whether texture semantics was better preserved after repeat-scaling than after zoom-scaling, but contrary to our hypothesis, we found that texture semantics was well-preserved after both types of enlargement. These findings suggest data visualizations that utilize geometric textures preserve their meaning across sizes, at least for the concepts and scales that we tested. A caveat is that here, we studied textures whose elements and spatial structure were discernible in the original and enlarged versions, but scaling may have a greater impact on semantics if changing size reveals/conceals details in texture elements or produces emergent features in spatial structure that were indiscernible at other scales.

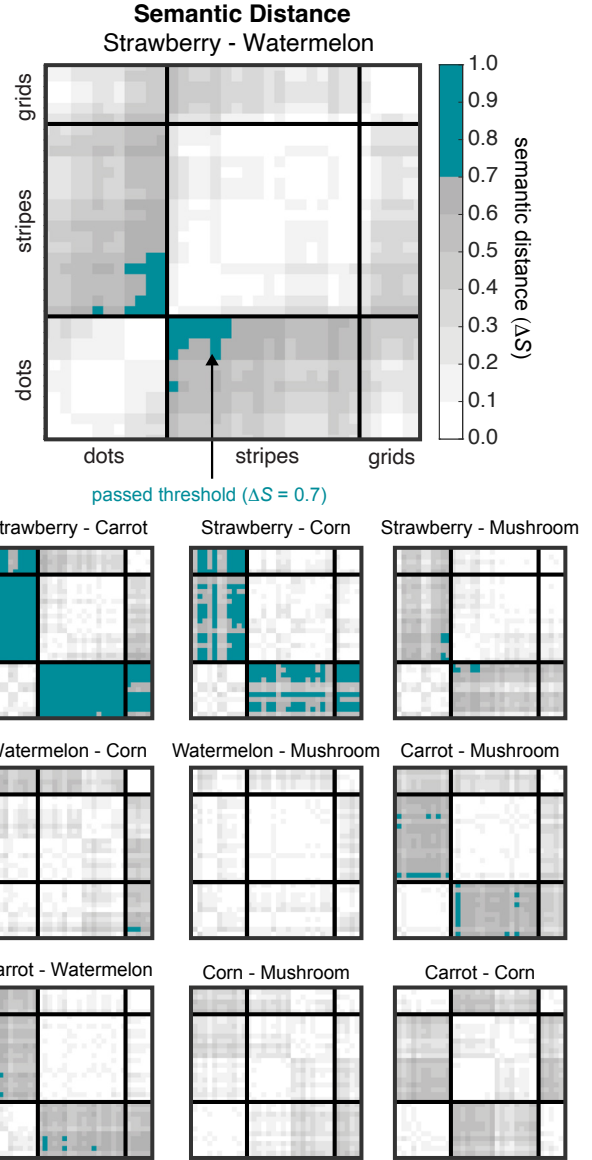


Figure 6: Semantic distances ΔS for each pair of 36 original textures within each concept pair. Darker colors indicate greater ΔS , with turquoise regions indicated $\Delta S \geq 0.7$. The axis labels indicate element shapes (dots, stripes, grids). See Supplementary Material Figure S.5 for a larger version that shows individual textures.

We also found that for the concepts we tested, textures were more likely to be semantically discriminable if they differed in element shape (dots, stripes, grids) than if they differed only in fill, orientation, or density. Questions remain whether this observation extends to other kinds of textures and concepts, especially for abstract concepts that lack resemblance for particular texture elements. Future work can address these questions toward establishing a comprehensive understanding of texture semantics for visual communication.

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